

TIME CAPSULE FOR COUPLES

A Digital Emotional Anchor & Conflict Resolution Platform for Partners

PROJECT LEAD
[Team Lead Name]

TARGET RELEASE
Q4 2026

DOCUMENT TYPE
Project Proposal

STATUS
Proposal Stage

1. EXECUTIVE SUMMARY

Time Capsule is an empathetic application designed to protect and support romantic relationships during periods of hardship, conflict, or distance. By encouraging partners to record video messages, write letters, and archive shared media during moments of high affection, the system seals these digital "capsules" to be dynamically unsealed when hardship occurs. It transforms past love into actionable grounding during vulnerable moments.

2. CORE FUNCTIONAL PILLARS

Contextual Triggers

Capsules unlock automatically during key events (anniversaries, periods of long distance) or emotional needs (e.g., "Open when we have our first fight").

Mutual Lock Protocol

Requires double confirmation from both partners before sealed memories can be authored, modified, or prematurely opened.

Guided Prompts

Psychology-backed templates designed to record healthy perspective (e.g., "What I love most about how we solve hard problems").

End-to-End Vault

Secure storage for high-vulnerability audio notes, letter drafts, photo albums, and video promises.

3. TARGET IMPACT & OBJECTIVES

85%

Conflict De-escalation Rate

100%

End-to-End Encrypted Data

2-Way

Shared Mutual Consent

4. SYSTEM ARCHITECTURE & TECH STACK

Architecture Layer	Selected Technology	Implementation Rationale
Mobile Client	React Native / Expo	Cross-platform native UI for seamless iOS & Android accessibility.
Backend API	Node.js (Fastify / Express)	Lightweight, asynchronous event handling for push triggers and auth.
Primary Database	PostgreSQL	Strict relational mapping for partner permissions and capsule metadata.
Media Storage	AWS S3 (Encrypted)	Secure object store for heavy media (audio, high-res video, photos).
Security & Auth	AES-256 + OAuth2 / JWT	Client-side encryption keys ensuring zero-knowledge cloud storage.

5. PRIVACY & SECURITY FRAMEWORK

Given the highly personal nature of romantic communications, **Time Capsule** operates under a strict Zero-Knowledge Architecture. Messages and videos are encrypted on-device before transmission, preventing any third-party or server admin access to raw emotional assets.

6. IMPLEMENTATION ROADMAP

Phase 1	Requirements & Wireframing (Weeks 1-3) User research, interface wireframes, database schema design, and cryptography spec finalize.
Phase 2	Core MVP Development (Weeks 4-9) Partner pairing logic, simple text/photo capsule creation, time-based unlocking triggers.
Phase 3	Media Integration & Emotion Triggers (Weeks 10-13) Audio/video recording integration, "Open when..." trigger workflows, end-to-end security audit.
Phase 4	Beta Testing & Launch (Weeks 14-16) Closed beta release to candidate couples, UX refinement, and official GitHub / Store deployment.